Istio Getting Started Lab Report

1. Objective

The objective of this lab was to install Istio on a Kubernetes cluster, deploy the Bookinfo sample application, expose the application to external traffic through the Kubernetes Gateway API, and observe service-to-service communication through the Kiali dashboard. According to the official Istio Getting Started guide, the main workflow includes installing Istio, installing the Gateway API CRDs, deploying the sample application, opening the application to outside traffic, and viewing the dashboard.

2. Environment

- Operating System: Windows
- Container Runtime: Docker Desktop
- Kubernetes Platform: Minikube
- Kubernetes CLI: kubectl
- Istio CLI: istioctl

A running Kubernetes cluster is required before installing Istio, and Minikube was used as the local Kubernetes environment in this experiment.

3. Experimental Procedure

Step 1. Verify the Istio installation

First, I verified that the Istio system components were running in the istio-system namespace. The output showed that istiod, istio-ingressgateway, and istio-egressgateway were all in the Running state, which confirmed that the Istio control plane and gateway components had been installed successfully.

（1） All Istio control plane and gateway components are running successfully in the istio-system namespace.

```
C:\WINDOWS\system32\cmd.
9ff2eb3d34c6e5500230961742b29dc715b5f1449f0e06620110e8b3": net/http: TLS handshake timeout)
Error: failed to install manifests: 2 errors occurred:
        * failed to wait for resource: resources not ready after 5m0s: context deadline exceeded
  Deployment/istio-system/istio-egressgateway (container failed to start: ImagePullBackOff: Back-off pulling image "dock
er.io/istio/proxyv2:1.29.2": ErrImagePull: Error response from daemon: Get "https://registry-1.docker.io/v2/": EOF)
        * failed to wait for resource: resources not ready after 5m0s: context deadline exceeded
  Deployment/istio-system/istio-ingressgateway (container failed to start: ImagePullBackOff: Back-off pulling image "doc
ker.io/istio/proxyv2:1.29.2": ErrImagePull: Get "https://registry-1.docker.io/v2/istio/proxyv2/manifests/sha256:3397bd41
9ff2eb3d34c6e5500230961742b29dc715b5f1449f0e06620110e8b3": net/http: TLS handshake timeout)


C:\Users\31018\Downloads\istioctl-1.29.2-win>docker pull docker.io/istio/proxyv2:1.29.2
1.29.2: Pulling from istio/proxyv2
e65b0115582d: Pull complete
ede6d5e95b67: Pull complete
689b91d88a0f: Pull complete
dec211a11132: Pull complete
2d099c62ee84: Pull complete
f354efe9a7ad: Pull complete
Digest: sha256:69ca4585993f4057861954f18be88df8268b713d114e7036bf9b185458c7a2c1
Status: Downloaded newer image for istio/proxyv2:1.29.2
docker.io/istio/proxyv2:1.29.2

C:\Users\31018\Downloads\istioctl-1.29.2-win>kubectl get pods -n istio-system
NAME                                    READY   STATUS    RESTARTS   AGE
istio-egressgateway-5bd755b557-bqpdq    1/1     Running   0          11m
istio-ingressgateway-6497cf79bc-h8fsn   1/1     Running   0          11m
istiod-75659b6dd4-7lslh                 1/1     Running   0          14m
```

（2）The Kubernetes Gateway API CRDs were installed successfully.

```
C:\Users\31018\Downloads\istioctl-1.29.2-win>kubectl kustomize "github.com/kubernetes-sigs/gateway-api/config/crd?ref=v1
.4.0" | kubectl apply -f -
customresourcedefinition.apiextensions.k8s.io/backendtlspolicies.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/gatewayclasses.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/gateways.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/grpcroutes.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/httproutes.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/referencegrants.gateway.networking.k8s.io created

C:\Users\31018\Downloads\istioctl-1.29.2-win>kubectl get crd gateways.gateway.networking.k8s.io
NAME                                 CREATED AT
gateways.gateway.networking.k8s.io   2026-04-16T03:48:20Z
```

(3) Deploy Application

The Bookinfo application was deployed by creating Pods, Deployments, and Services.

```
C:\Users\31018\Downloads\istioctl-1.29.2-win>cd C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2

C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl apply -f samples/bookinfo/platform/kube/b
ookinfo.yaml
service/details created
serviceaccount/bookinfo-details created
deployment.apps/details-v1 created
service/ratings created
serviceaccount/bookinfo-ratings created
deployment.apps/ratings-v1 created
service/reviews created
serviceaccount/bookinfo-reviews created
deployment.apps/reviews-v1 created
deployment.apps/reviews-v2 created
deployment.apps/reviews-v3 created
service/productpage created
serviceaccount/bookinfo-productpage created
deployment.apps/productpage-v1 created
```

(4) Create Gateway

A Gateway resource was created to allow external traffic to access the Bookinfo application.

```
C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl apply -f samples/bookinfo/gateway-api/boo
kinfo-gateway.yaml
gateway.gateway.networking.k8s.io/bookinfo-gateway created
httproute.gateway.networking.k8s.io/bookinfo created
```

(5) Check Gateway Status

The Gateway status initially showed **False**, which indicates:

- The Gateway had not yet been fully activated
- It was not connected to the ingress gateway
- External access was not enabled (Minikube tunnel not started)

```
C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl get gateway
NAME               CLASS   ADDRESS   PROGRAMMED   AGE
bookinfo-gateway   istio             False        14s

C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl get httproute
NAME       HOSTNAMES   AGE
bookinfo               31s
```

(6) Obtain External Access Address

This step is critical. For Minikube, it is necessary to run the following command in a separate terminal:

minikube tunnel

The Minikube tunnel was enabled to expose the gateway service externally.

An external access path for the Bookinfo application was successfully created through the gateway.

```
C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>minikube tunnel
* 隧道成功启动

* 注意：请不要关闭此终端，因为此进程必须保持活动状态才能访问隧道......

! 在 Windows 上使用 v8.1以上版本的OpenSSH客户端，访问 1024 以下端口可能会失败。更多信息请参阅：https://minikube.sigs.k8s
.io/docs/handbook/accessing/#access-to-ports-1024-on-windows-requires-root-permission
* 为服务 bookinfo-gateway-istio 启动隧道。
! 在 Windows 上使用 v8.1以上版本的OpenSSH客户端，访问 1024 以下端口可能会失败。更多信息请参阅：https://minikube.sigs.k8s
.io/docs/handbook/accessing/#access-to-ports-1024-on-windows-requires-root-permission
* 为服务 istio-ingressgateway 启动隧道。
```

An external access path for the Bookinfo application was created through the gateway.

```
C:\Users\31018>kubectl get svc -n istio-system
NAME                   TYPE           CLUSTER-IP      EXTERNAL-IP   PORT(S)
  AGE
istio-egressgateway    ClusterIP      10.102.22.26    <none>        80/TCP,443/TCP
  52m
istio-ingressgateway   LoadBalancer   10.97.171.108   127.0.0.1     15021:31807/TCP,80:31713/TCP,443:30871/TCP,31400:306
55/TCP,15443:32527/TCP   52m
istiod                 ClusterIP      10.97.120.104   <none>        15010/TCP,15012/TCP,443/TCP,15014/TCP
  55m
```

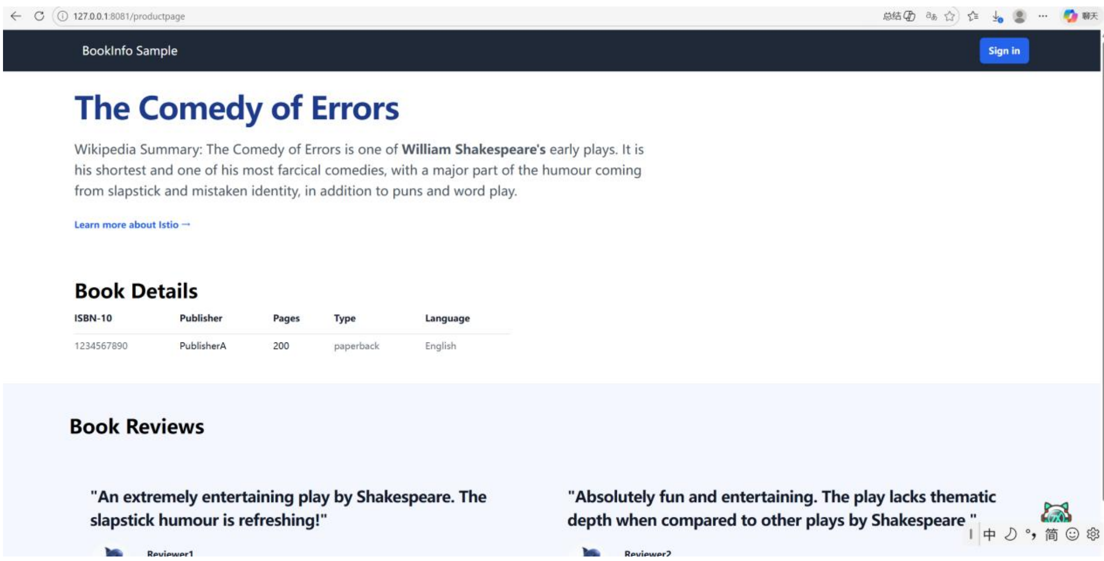
（7）The application was verified both through a web browser and via curl, ensuring that the service mesh routing is functioning correctly.

```
C:\Users\31018>kubectl get gateway
NAME               CLASS    ADDRESS     PROGRAMMED   AGE
bookinfo-gateway   istio    127.0.0.1   True         18m

C:\Users\31018>kubectl get httproute
NAME       HOSTNAMES   AGE
bookinfo               18m

C:\Users\31018>kubectl get pods
NAME                                        READY   STATUS    RESTARTS   AGE
bookinfo-gateway-istio-6f4b86fd99-65vtp     1/1     Running   0          18m
details-v1-764c46cfdb-jfwwj                 1/1     Running   0          22m
productpage-v1-85664dccbc-xqq9x             1/1     Running   0          22m
ratings-v1-779fdc7f86-qdbpb                 1/1     Running   0          22m
reviews-v1-85964f9f98-95jd9                 1/1     Running   0          22m
reviews-v2-6f7fbdc6fd-fzndq                 1/1     Running   0          22m
reviews-v3-689b477554-k2lcj                 1/1     Running   0          22m

C:\Users\31018>
```

The Bookinfo application was successfully accessed through the Istio gateway using curl, and the HTML response confirms correct service routing.

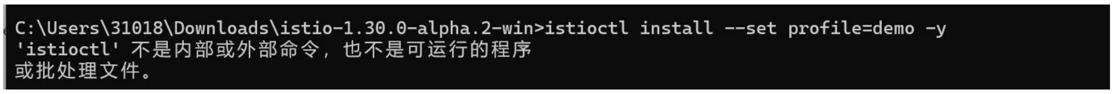
```
C:\Users\31018\Downloads>curl http://127.0.0.1:8081/productpage

<meta charset="utf-8">
<meta http-equiv="X-UA-Compatible" content="IE=edge">
<meta name="viewport" content="width=device-width, initial-scale=1.0">


<title>Simple Bookstore App</title>

<script src="static/tailwind/tailwind.css"></script>
<script type="text/javascript">
  window.addEventListener("DOMContentLoaded", (event) => {
    const dialog = document.querySelector("dialog");
    const showButton = document.querySelector("#sign-in-button");
    const closeButton = document.querySelector("#close-dialog");

    if (showButton) {
      showButton.addEventListener("click", () => {
        dialog.showModal();
      });
    }

    if (closeButton) {
      closeButton.addEventListener("click", () => {
        dialog.close();
      });
```

The Bookinfo application was successfully accessed through the Istio gateway, displaying product details and user reviews



4. Troubleshooting

（1）The system could not recognize the istioctl command.
The istioctl.exe was not added to the system PATH.



```
C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win>istioctl install --set profile=demo -y
'istioctl' 不是内部或外部命令，也不是可运行的程序
或批处理文件。
```

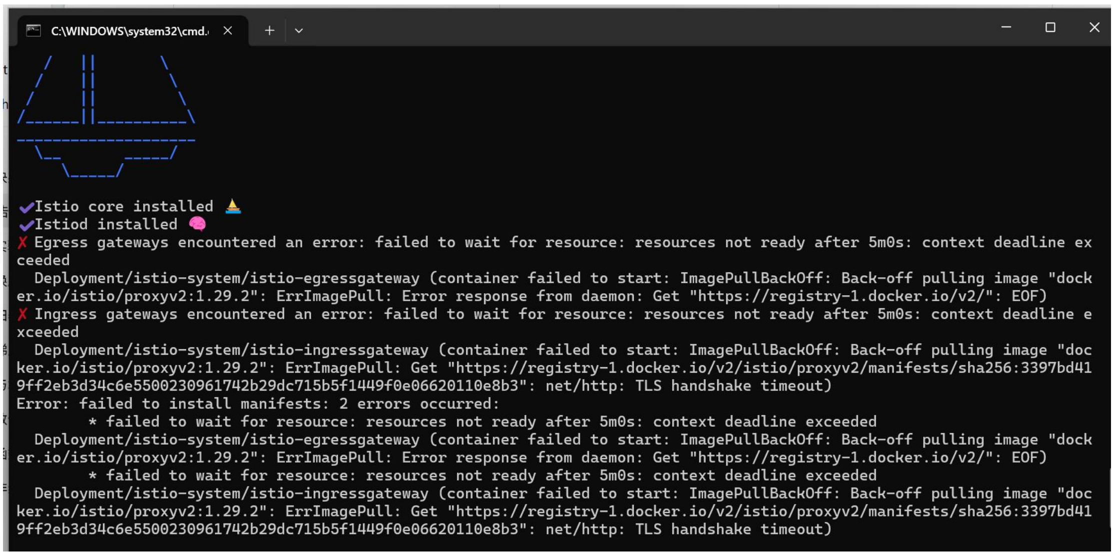
Solution:

Run `.\istioctl.exe` inside the `bin` directory

Or add the `bin` directory to the system PATH

```
C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2\bin>istioctl install --set profile=demo -y
```

（2）Istio core and control plane were installed, but the gateway components failed due to image pull timeout.



```
C:\WINDOWS\system32\cmd.

✔Istio core installed ⛵
✔Istiod installed 🧠
✗ Egress gateways encountered an error: failed to wait for resource: resources not ready after 5m0s: context deadline exceeded
  Deployment/istio-system/istio-egressgateway (container failed to start: ImagePullBackOff: Back-off pulling image "docker.io/istio/proxyv2:1.29.2": ErrImagePull: Error response from daemon: Get "https://registry-1.docker.io/v2/": EOF)
✗ Ingress gateways encountered an error: failed to wait for resource: resources not ready after 5m0s: context deadline exceeded
  Deployment/istio-system/istio-ingressgateway (container failed to start: ImagePullBackOff: Back-off pulling image "docker.io/istio/proxyv2:1.29.2": ErrImagePull: Get "https://registry-1.docker.io/v2/istio/proxyv2/manifests/sha256:3397bd419ff2eb3d34c6e5500230961742b29dc715b5f1449f0e06620110e8b3": net/http: TLS handshake timeout)
Error: failed to install manifests: 2 errors occurred:
        * failed to wait for resource: resources not ready after 5m0s: context deadline exceeded
  Deployment/istio-system/istio-egressgateway (container failed to start: ImagePullBackOff: Back-off pulling image "docker.io/istio/proxyv2:1.29.2": ErrImagePull: Error response from daemon: Get "https://registry-1.docker.io/v2/": EOF)
        * failed to wait for resource: resources not ready after 5m0s: context deadline exceeded
  Deployment/istio-system/istio-ingressgateway (container failed to start: ImagePullBackOff: Back-off pulling image "docker.io/istio/proxyv2:1.29.2": ErrImagePull: Get "https://registry-1.docker.io/v2/istio/proxyv2/manifests/sha256:3397bd419ff2eb3d34c6e5500230961742b29dc715b5f1449f0e06620110e8b3": net/http: TLS handshake timeout)
```

Solution：Manually pull the required image:docker pull docker.io/istio/proxyv2:1.29.2

(3) Apache Default Page Appears in Browser

Problem:
The browser displayed the Apache default page instead of the application.

Cause:
Local Apache server was occupying port 80.

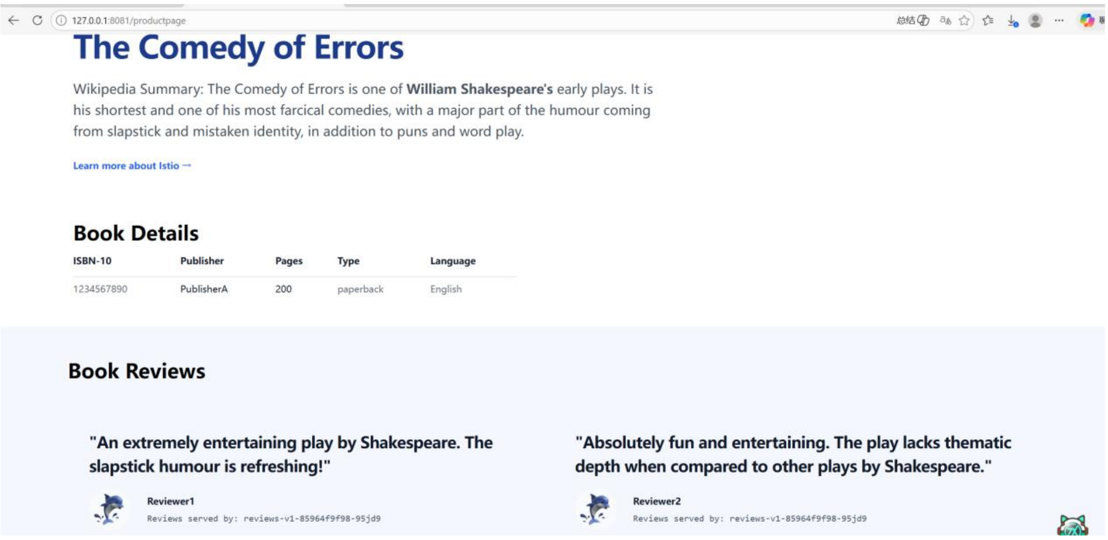
Solution:
Use port forwarding to map a different port:

kubectl port-forward svc/bookinfo-gateway-istio 8081:80

Access via:

http://127.0.0.1:8081/productpage

http://127.0.0.1:<port>/productpage



5. Conclusion & Reflection

Through this experiment, I successfully learned how to install Istio in a Kubernetes environment and deploy a microservices-based application using the Bookinfo example. I gained hands-on experience with configuring the Gateway API, enabling external access using Minikube tunnel, and verifying service communication through browser and curl requests. During the process, I encountered several issues such as command recognition errors, image pull timeouts, and port conflicts, which helped me better understand the underlying system configuration and troubleshooting methods. Overall, this lab improved my understanding of service mesh architecture and strengthened my practical skills in Kubernetes and Istio deployment.